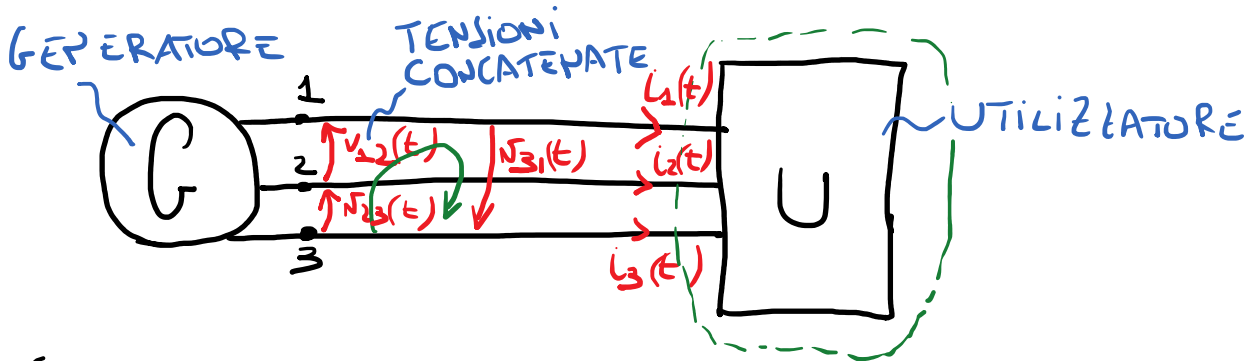


SISTEMI TRIFASE

venerdì 13 maggio 2022 09:16



$$\begin{cases} LKT: v_{12}(t) + v_{23}(t) + v_{31}(t) = 0 \\ LKC: i_1(t) + i_2(t) + i_3(t) = 0 \end{cases}$$

Nella realtà: **SISTEMI TRIFASE SIMMETRICI**

- sinusoidali

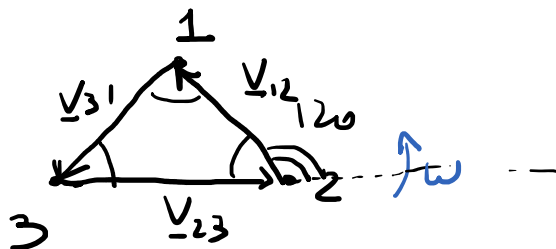
- ampiezze uguali: $\hat{v}_{12} = \hat{v}_{23} = \hat{v}_{31} = \hat{v}$

$\Downarrow S[\]$

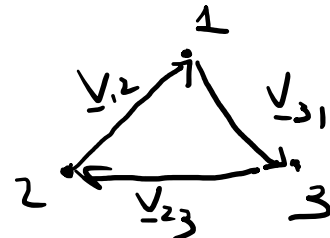
$$|v_{12}| = |v_{23}| = |v_{31}| = |v|$$

$$\begin{cases} v_{12} + v_{23} + v_{31} = 0 \\ |v_{12}| = |v_{23}| = |v_{31}| = v \end{cases}$$

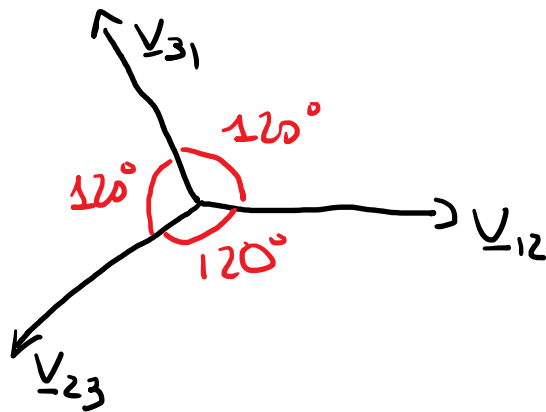
TERNA DIRETTA



TERNA INVERSA

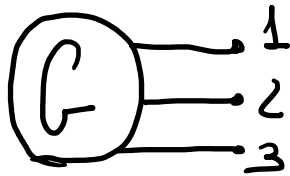


Ridipingo la Terna:

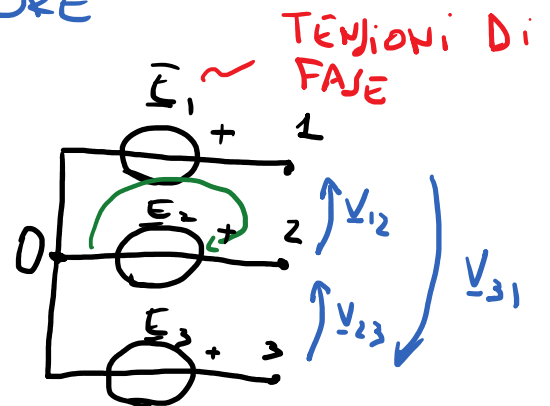


$$\begin{cases} \underline{V}_{12} = V e^{j0} \\ \underline{V}_{23} = V e^{-j\frac{2}{3}\pi} = \underline{V}_{12} e^{-j\frac{2}{3}\pi} = \underline{V}_{23} \\ \underline{V}_{31} = V e^{j\frac{2}{3}\pi} = \underline{V}_{12} e^{j\frac{2}{3}\pi} = \underline{V}_{31} \end{cases}$$

MODELLO DEL GENERATORE

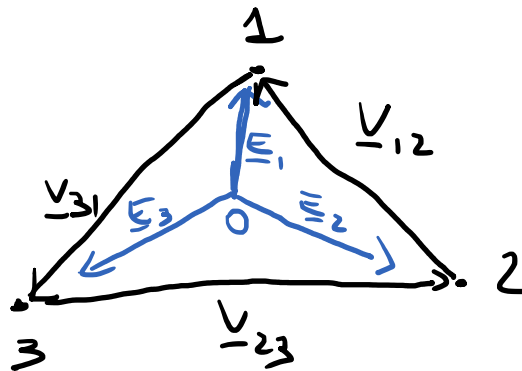


$\equiv$



$$LKT: \underline{E}_1 - \underline{V}_{12} - \underline{E}_2 = 0 \rightarrow \begin{cases} \underline{V}_{12} = \underline{E}_1 - \underline{E}_2 \\ \underline{V}_{23} = \underline{E}_2 - \underline{E}_3 \\ \underline{V}_{31} = \underline{E}_3 - \underline{E}_1 \end{cases}$$

TERNA DIRETTA



$\underline{E}$ sono una terna simmetrica:

$$|\underline{E}_1| = |\underline{E}_2| = |\underline{E}_3| = \bar{E}$$

$$V = \sqrt{3} E$$

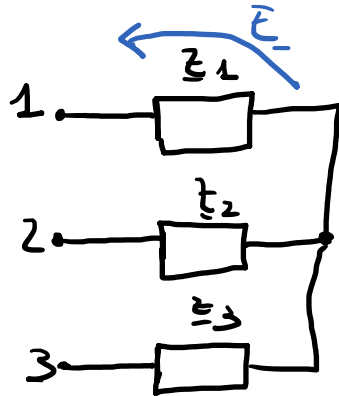
In Italia:

$$V = 400V$$

$$E = 230V$$

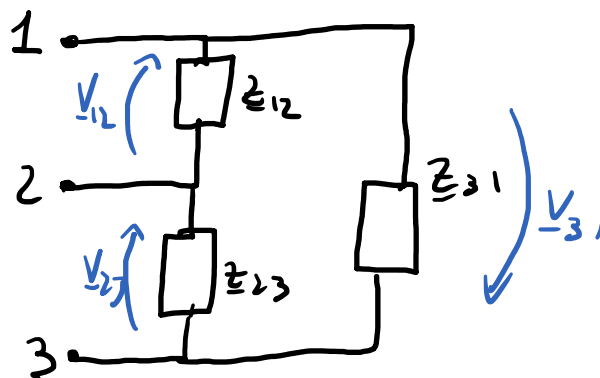
CARICO

- Connessione a stella (Y)



=> le impedenze vedono la tensione di Fase

- Connessione a triangolo (D)



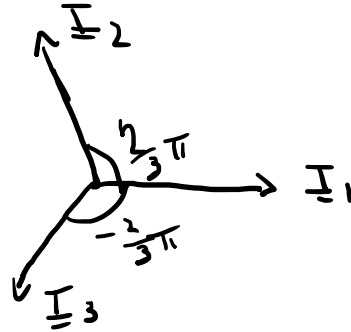
=> le impedenze vedono la tensione lineare ($V = \sqrt{3} E$)

- Se le impedenze sono uguali:

$$z_1 = z_2 = z_3 = z \rightarrow |I_1| = |I_2| = |I_3| = I$$

CARICO
EQUILIBRATO

⇓
TERZA SINDRICA



$$\underline{N} = \underline{E}_{10} \underline{I}_1^* + \underline{E}_{20} \underline{I}_2^* + \underline{E}_{30} \underline{I}_3^* = 3 \underline{E} \underline{I}^*$$

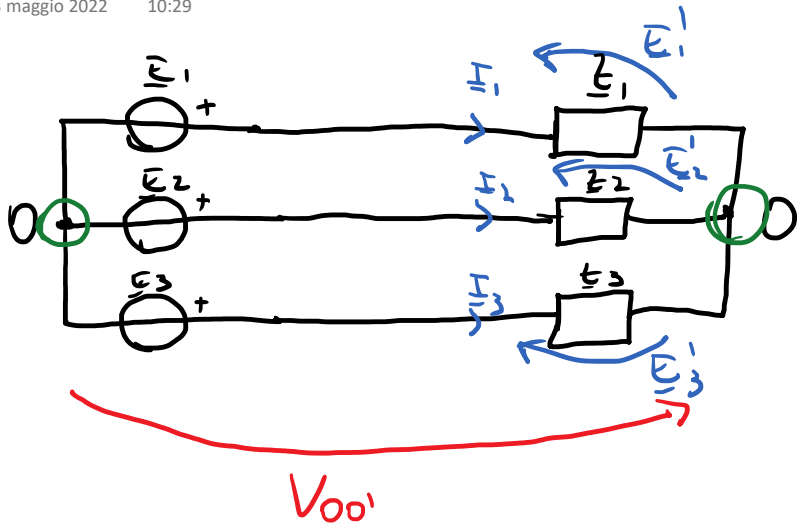
$$= 3 EI (\cos \varphi + j \sin \varphi) = \sqrt{3} \sqrt{3} EI (\cos \varphi + j \sin \varphi)$$

$$= \sqrt{3} V I (\cos \varphi + j \sin \varphi)$$

$$\begin{cases} P = \sqrt{3} V I \cos \varphi \\ Q = \sqrt{3} V I \sin \varphi \end{cases}$$

SISTEMA TRIFASE A 4 CONDUTTORI

venerdì 13 maggio 2022 10:29



Perfiamo con
una terra
simmetrica

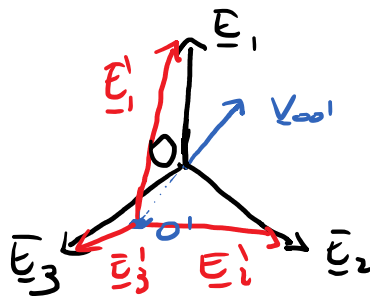
$$\underline{E}_1' = ?$$

$$\underline{E}_2' = ?$$

$$\underline{E}_3' = ?$$

$$\begin{cases} \underline{E}_1' = \underline{E}_1 - V_{00'} \\ \underline{E}_2' = \underline{E}_2 - V_{00'} \\ \underline{E}_3' = \underline{E}_3 - V_{00'} \end{cases}$$

Supponiamo una $V_{00'}$ ↗



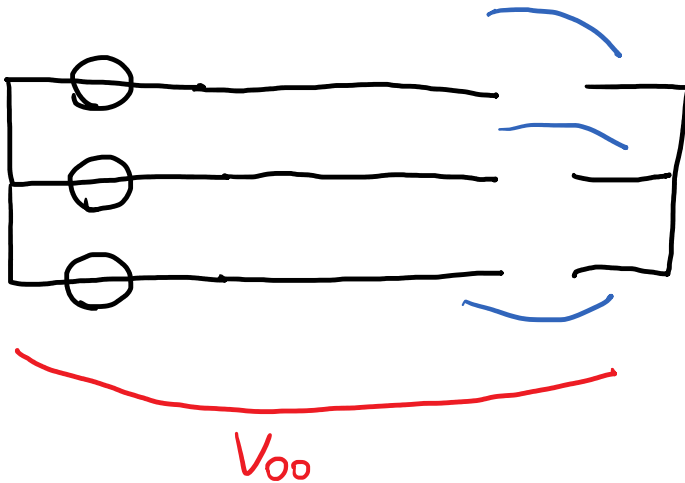
$$\left. \begin{aligned} |\underline{E}_1'| &> |\underline{E}_1| \\ |\underline{E}_2'| &> |\underline{E}_2| \end{aligned} \right\} \text{SOVRATON.}$$

$$|\underline{E}_3'| < |\underline{E}_3| \rightarrow \text{PALFUNZIONAMENTO}$$

⇓
 $\underline{E}'$ NON SIMMETRICHE

Vorremmo che $V_{\infty'} = 0 \Rightarrow \begin{cases} \underline{E}_1' = \underline{E}_1 \\ \underline{E}_2' = \underline{E}_2 \\ \underline{E}_3' = \underline{E}_3 \end{cases}$

Calcoliamo $V_{\infty'}$



$$\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0$$

$$\frac{\underline{E}_1 - V_{\infty'}}{\underline{z}_1} + \frac{\underline{E}_2 - V_{\infty'}}{\underline{z}_2} + \frac{\underline{E}_3 - V_{\infty'}}{\underline{z}_3} = 0$$

$$\frac{\underline{E}_1}{\underline{z}_1} - \frac{V_{\infty'}}{\underline{z}_1} + \frac{\underline{E}_2}{\underline{z}_2} - \frac{V_{\infty'}}{\underline{z}_2} + \frac{\underline{E}_3}{\underline{z}_3} - \frac{V_{\infty'}}{\underline{z}_3} = 0$$

$$V_{\infty'} = \frac{\underline{E}_1/\underline{z}_1 + \underline{E}_2/\underline{z}_2 + \underline{E}_3/\underline{z}_3}{1/\underline{z}_1 + 1/\underline{z}_2 + 1/\underline{z}_3}$$

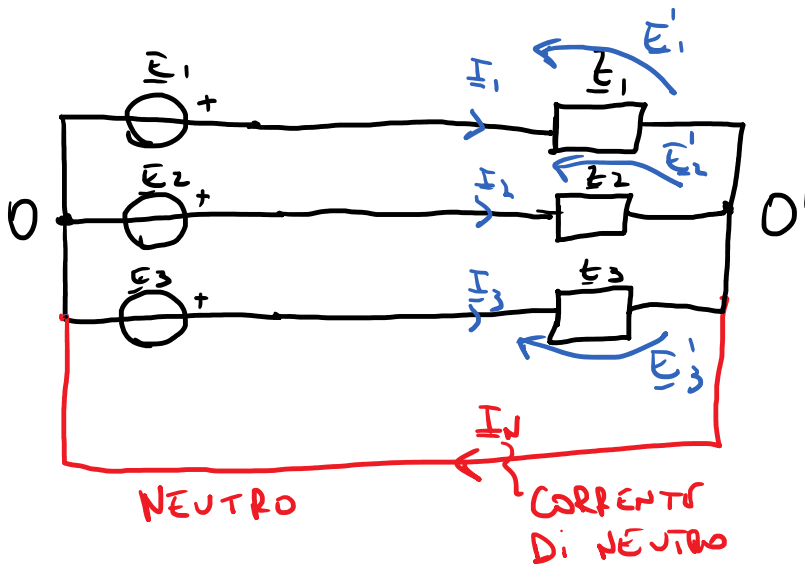
- se il carico è equilibrato ($\underline{z}_1 = \underline{z}_2 = \underline{z}_3 = \underline{z}$)

$$V_{ao'} = \frac{\frac{1}{\underline{z}} (\underline{E}_1 + \underline{E}_2 + \underline{E}_3)}{3/\underline{z}} = 0 \quad V \Rightarrow \begin{aligned} \underline{E}_1' &= \underline{E}_1 \\ \underline{E}_2' &= \underline{E}_2 \\ \underline{E}_3' &= \underline{E}_3 \end{aligned}$$

- se il carico è squilibrato: $V_{ao'} \neq 0$

Tanto più il carico è squilibrato tanto più $V_{ao'} \neq 0$

$\Rightarrow$ Fortemente $V_{ao'}$ a 0 $\rightarrow$ caso caso 0 e 0'



$$\underline{I}_N = \underline{I}_1 + \underline{I}_2 + \underline{I}_3 = \frac{\underline{E}_1}{\underline{z}_1} + \frac{\underline{E}_2}{\underline{z}_2} + \frac{\underline{E}_3}{\underline{z}_3}$$

$\underline{I}_N$ sarà tanto più alta tanto più il carico è squilibrato

- Il nostro fa in modo di avere una forma simmetrica sul carico